MODULE 27 · DOMAIN 3 - COST-OPTIMIZED ARCHITECTURE DESIGN

AI/ML Cost Optimisation

Choose the tier and model each request deserves, batch what can wait, cache what repeats, and route inference without paying to move data.

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admin@sthithapragna.com

Most AI spend is **tokens times price**. Send fewer tokens, to a cheaper model, at a cheaper tier.

Bedrock inference options and cost

OPTION	FITS	COST EFFECT
On-demand, Standard tier	Interactive, everyday traffic	Baseline per-token price
Priority tier	Latency-critical, customer-facing	Premium for faster output
Flex tier	Agentic or background, can wait	Discount for lower priority
Batch inference	Large jobs, results later	About half the on-demand price
Provisioned throughput	Steady high volume, custom models	Pay for model units by the hour

ON THE EXAM

Results needed tomorrow, not now: batch. Can wait seconds longer: Flex. Must be fastest: Priority.

Prompt caching

- **Caches a repeated prompt prefix**, such as a long system prompt or document.
- **Cache reads** are billed far below normal input; up to 90% lower cost.
- **TTL** usually 5 minutes, reset on each hit; some models offer 1 hour.
- **Prefix must be identical**; put changing content after the checkpoint.
- **On-demand only**; not available with batch inference.

> ON THE EXAM
>
> The same long context sent with every request: prompt caching.

Prompt routing or distillation?

INTELLIGENT PROMPT ROUTING

Pick a model per request

- Router chooses within a model family
- Simple prompts go to the smaller model
- Up to 30% lower cost
- No training needed

MODEL DISTILLATION

Train a small model to copy a big one

- Teacher model generates training data
- Student model learns one task
- Much faster and cheaper to run
- Best for narrow, high-volume tasks

THE TIEBREAKER

Mixed difficulty, no training effort: routing. One narrow task at huge volume: distillation.

SageMaker AI inference options

OPTION	FITS
Real-time endpoint	Steady, low-latency traffic
Serverless inference	Intermittent traffic; scales to zero; cold starts
Asynchronous inference	Large payloads, long processing; scales to zero
Batch transform	Whole datasets offline; no endpoint kept
Multi-model and inference components	Many models sharing one endpoint's instances

ON THE EXAM

Idle most of the day and cold starts acceptable: serverless. Many small models: share an endpoint.

Training cost levers

- **Managed Spot training** with checkpoints, for large discounts.
- **SageMaker AI Savings Plans** for steady training and inference.
- **Right-size instances**; stop idle notebooks and endpoints.
- **Capacity Blocks or HyperPod** for long, large GPU runs.
- **Fine-tune or distil** a small model instead of always calling a large one.

> ON THE EXAM
>
> Checkpointed training jobs: managed Spot. Idle endpoints: Compute Optimizer finds them.

Calling another Region, or cross-Region inference?

APPLICATION CALLS A REMOTE REGION

You move the data

- Inter-Region data transfer charges
- Added latency on every call
- Your code manages failover
- May break residency rules

CROSS-REGION INFERENCE PROFILE

Bedrock routes for you

- Call your local Region only
- Bedrock routes over the AWS network
- Priced at the source Region
- Geographic or global profiles

THE TIEBREAKER

More capacity without paying to move prompts between Regions: a cross-Region inference profile.

Reading the scenario

WHEN THE QUESTION SAYS	IT IS TESTING
Millions of prompts, results by morning	Batch inference
Same long context every request	Prompt caching
Mixed easy and hard questions	Intelligent prompt routing
One narrow task, huge volume	Model distillation
Model idle most of the day	SageMaker serverless inference
Calling another Region for capacity	Cross-Region inference profile

ON THE EXAM

Find the waste: repeated tokens, oversized model, wrong tier, idle endpoint, or data moved needlessly.

> Each night, Meridian Freight summarises two million customs documents with a Bedrock model. Summaries are read by analysts the next morning.
>
> The job currently runs on-demand and is the largest line on the AI bill.

What should the solutions architect recommend?

A Switch the job to the Priority service tier

B Buy provisioned throughput for the overnight window

C Submit the documents as a Bedrock batch inference job

D Enable prompt caching on each request

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Priority tier

Pays more for speed nobody needs overnight.

B Provisioned throughput

Hourly commitment for a job batch handles more cheaply.

C Batch inference

Asynchronous, large-scale, at a large discount to on-demand.

D Prompt caching

Each document differs, and caching is not available in batch.

THE SIGNAL

Large volume, results later: batch inference.

> Meridian Freight's customs assistant sends a 25,000-token system prompt, including the full tariff guide, with every question. Questions arrive several times a minute all day.
>
> Most of each request's input cost is the unchanging guide.

What should the solutions architect recommend?

A Use prompt caching, with a cache checkpoint after the system prompt and guide

B Move the assistant to batch inference

C Switch to a larger model with a bigger context window

D Buy provisioned throughput

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Prompt caching

The guide is read from cache at a fraction of the input price.

B Batch

An interactive assistant cannot wait for a batch job.

C Larger model

Costs more per token, and the context already fits.

D Provisioned throughput

Commits spend without removing the repeated tokens.

THE SIGNAL

An identical long prefix, requested often enough to stay warm: prompt caching.

> Meridian Freight's support assistant uses a large model for every question. Analysis shows most questions are simple tracking lookups, while some are complex claims disputes.
>
> The team wants lower cost without building or training anything.

What should the solutions architect recommend?

A Distil a small model from the large one

B Switch every question to the smallest model

C Use the Priority tier

D Use Bedrock intelligent prompt routing across a small and large model in the same family

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Distillation

Involves training, which the team ruled out.

B Smallest model for all

Cheaper, but complex disputes would suffer.

C Priority tier

Costs more.

D Prompt routing

Simple questions go to the small model, hard ones to the large.

THE SIGNAL

Mixed difficulty, no training: intelligent prompt routing.

> Meridian Freight extracts the same 12 fields from millions of bills of lading a day. A large model does this accurately, but cost and latency are too high.
>
> The team accepts a small accuracy loss and is willing to invest in a one-off customisation.

What should the solutions architect recommend?

A Enable prompt caching for each document

B Use Bedrock Model Distillation to train a smaller model on the large model's outputs for this task

C Use intelligent prompt routing

D Buy provisioned throughput for the large model

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Prompt caching

Each document differs; little to cache.

B Distillation

A small model learns this one task, far cheaper and faster.

C Prompt routing

Every request is the same difficulty; routing adds little.

D Provisioned throughput

Still runs the expensive model.

THE SIGNAL

One narrow task, huge volume, customisation acceptable: distillation.

> Meridian Freight hosts a damage-photo classification model on a SageMaker AI real-time endpoint. Requests arrive in bursts a few times a day; the endpoint is idle over 90% of the time.
>
> A delay of a few seconds on the first request after idle is acceptable.

What should the solutions architect recommend?

A. Move the model to SageMaker AI Serverless Inference

B. Buy a SageMaker AI Savings Plan for the endpoint

C. Move to a larger instance type for faster responses

D. Run a batch transform job every minute

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Serverless inference
Scales to zero when idle; cold starts are accepted.

B Savings Plan
Discounts an endpoint that is idle 90% of the time.

C Larger instance
More cost for idle capacity.

D Batch every minute
Clumsy, and not real-time request handling.

THE SIGNAL

Idle most of the time, cold starts acceptable: serverless inference.

> Meridian Freight's application in eu-west-1 calls a Bedrock model endpoint in us-east-1 directly, because it needed more capacity at peak. The team now sees inter-Region data transfer charges and higher latency.
>
> Processing anywhere within the EU is acceptable.

What should the solutions architect recommend?

A Move the application to us-east-1

B Buy provisioned throughput in us-east-1

C Add a VPC peering connection between the Regions

D Call Bedrock in eu-west-1 using the EU geographic cross-Region inference profile

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Move to us-east-1

A large migration, and moves data out of the EU.

B Provisioned in us-east-1

Keeps the cross-Region calls and their charges.

C VPC peering

Still inter-Region traffic, still charged.

D EU inference profile

Local calls; Bedrock adds EU capacity, priced at the source Region.

THE SIGNAL

Need more capacity without paying to move prompts: cross-Region inference from the local Region.

What to carry out of this module

- **Tokens times price:** fewer tokens, cheaper model, cheaper tier.
- **Batch** for later; **Flex** for patient; **Priority** only for latency-critical.
- **Caching** for repeated prefixes; **routing** or **distillation** for cheaper models.
- **SageMaker:** serverless or async for idle endpoints, Spot for training.
- **Cross-Region inference** instead of calling remote Regions yourself.

> ON THE EXAM
>
> Find the AI waste, then use the feature built for it.